

In [44]: `import pandas as pd`

```
# Load all available indicators
df = pd.read_csv('../data/who_obesity_data.csv')
df.shape
```

C:\Users\Jack\AppData\Local\Temp\ipykernel\_21604\4030832271.py:5: DtypeWarning: Columns (0,1,2,4,5,6,7,8,9,17,18,25,26,27) have mixed types. Specify dtype option on import or set low\_memory=False.

```
df = pd.read_csv('../data/who_obesity_data.csv')
```

Out[44]: (3961866, 28)

In [63]: `# Calculate total number of unique sub-dimensions in Dim1, Dim2, and Dim3`

```
subdim_cols = ['Dim1', 'Dim2', 'Dim3']
unique_subdims = set()
for col in subdim_cols:
    unique_subdims.update(df[col].dropna().unique())
print(f"Total unique sub-dimensions: {len(unique_subdims)}")

# Calculate total number of unique sub-dimensions in Dim2 only
num_unique_dim2 = df['Dim2'].dropna().nunique()
print(f"Total unique sub-dimensions in Dim2: {num_unique_dim2}")
```

Total unique sub-dimensions: 5303

Total unique sub-dimensions in Dim2: 119

In [55]: `# Examine dimension values`

```
print("Dim1 (Sex) Value counts:")
print(df['Dim1'].value_counts(dropna=False))
print("\n" + "="*50 + "\n")

print("Dim2 (Categories) - Value counts:")
print(df['Dim2'].value_counts(dropna=False))
print("\n" + "="*50 + "\n")

print("Dim1Type values:")
print(df['Dim1Type'].value_counts(dropna=False))
print("\n" + "="*50 + "\n")

print("Dim2Type values:")
print(df['Dim2Type'].value_counts(dropna=False))
```

Dim1 (Sex) Value counts:

|                        |        |
|------------------------|--------|
| Dim1                   |        |
| SEX_FMLE               | 907147 |
| SEX_MLE                | 905115 |
| SEX_BT SX              | 602371 |
| NaN                    | 394149 |
| RESIDENCEAREATYPE_TOTL | 131667 |

...

|                                            |   |
|--------------------------------------------|---|
| AGEGROUP_YEARS30-39                        | 2 |
| AGEGROUP_YEARS10-14                        | 2 |
| AGEGROUP_YEARS30-49                        | 2 |
| AGEGROUP_YEARS10-19                        | 2 |
| AMRGLASSCATEGORY_AMRGLASS_PROVIDEDTOLABS07 | 2 |

Name: count, Length: 5070, dtype: int64

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Dim2 (Categories) - Value counts:

|                       |         |
|-----------------------|---------|
| Dim2                  |         |
| NaN                   | 1946809 |
| AGEGROUP_YEARS18-PLUS | 305859  |
| AGEGROUP_AGEAll       | 136244  |
| AGEGROUP_YEARS10-19   | 113724  |
| AGEGROUP_YEARS15-29   | 89661   |

...

|                                             |    |
|---------------------------------------------|----|
| ASSISTIVETECHSOURCE_ASSISTIVETECH_Private   | 52 |
| ASSISTIVETECHSOURCE_ASSISTIVETECH_Self-made | 52 |
| ASSISTIVETECHSOURCE_ASSISTIVETECH_NGO       | 52 |
| ASSISTIVETECHSOURCE_ASSISTIVETECH_Public    | 52 |
| EMFFREQUENCY_EMFSTATIC                      | 49 |

Name: count, Length: 120, dtype: int64

=====

Dim1Type values:

|                   |         |
|-------------------|---------|
| Dim1Type          |         |
| SEX               | 2414633 |
| NaN               | 394149  |
| RESIDENCEAREATYPE | 385514  |
| TOBACCO_INDICATOR | 376244  |
| ALCOHOLTYPE       | 138284  |
| DHSMICSGEOREGION  | 40410   |
| WEALTHDECILE      | 40351   |
| AGEGROUP          | 29580   |
| WEALTHQUINTILE    | 25215   |

|                          |       |
|--------------------------|-------|
| EDUCATIONLEVEL           | 18047 |
| SEVERITY                 | 16160 |
| TOBACCO_NICOTINE_PRODUCT | 13065 |
| RSUDREP                  | 7760  |
| HOUSEHOLD_COMP_BY_AGE    | 6902  |
| HOUSEHOLDWEALTH          | 5518  |
| RSUDSPECIFICPOPULATION   | 5432  |
| RSUDTREATMENTLOCATION    | 4656  |
| RSUDHWF                  | 3880  |
| BEVERAGETYPE             | 3854  |
| SCSUDSURVEYQUESTIONS     | 3335  |
| PUBLICPLACE              | 2304  |
| PROGRAMME                | 1940  |
| TB_TREATHISTORY          | 1899  |
| DRUG                     | 1882  |
| ADVERTISINGTYPE          | 1626  |
| ENVCAUSE                 | 1552  |
| RSUDPREVENTIONPROGRAMME  | 1552  |
| SPECIALPOPULATION        | 1305  |
| POLICYSECTOR             | 1237  |
| GOEQUESTION              | 1223  |
| ARCHIVE                  | 1153  |
| WEALTHTERCILE            | 1048  |
| BACGROUP                 | 1044  |
| SUBSTANCETYPEDISORDER    | 1022  |
| GROUP                    | 970   |
| NATIONALSYSTEMTYPE       | 878   |
| COORDINATINGENTITIES     | 840   |
| RSUDTREATMENT            | 776   |
| OPENACCESSSERVICE        | 582   |
| NGO                      | 544   |
| GOVERNMENTBENEFIT        | 498   |
| POLICYFRAMEWORK          | 498   |
| ASSISTIVETECHSUBQUESTION | 490   |
| RSUDMONITORING           | 388   |
| DRIVERTYPE               | 350   |
| ASSISTIVETECHPRODUCT     | 312   |
| HARMANDCONSEQUENCE       | 275   |
| EMFEXPOSED               | 232   |
| CONSUMPTIONTYPE          | 126   |
| SOCIALCOSTTYPE           | 108   |
| TB_TREATMENTTYPE         | 92    |
| LEGISLATIONTYPE          | 55    |
| POLICYADOPTIONLEVEL      | 47    |
| AMRGLASSCATEGORY         | 24    |

PRICE 5

Name: count, dtype: int64

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Dim2Type values:

Dim2Type

NaN 1946809

AGEGROUP 1582086

HOUSEHOLDCOOKINGFUEL 204408

GHECAUSE 135720

TOBACCO\_NICOTINE\_PRODUCT 36855

ENVCAUSE 16722

ARCHIVE 16652

PRICE 12675

POP\_TYPE 6658

BEVERAGE 2340

TB\_CASETYPE 397

ASSISTIVETECHSOURCE 312

EMFFREQUENCY 232

Name: count, dtype: int64

```
In [56]: # Check all Dim1Type categories and their top values
print("=" * 60)
print("DIM1 TYPES AND THEIR VALUES")
print("=" * 60)

for dim_type in df['Dim1Type'].dropna().unique():
    dim_values = df[df['Dim1Type'] == dim_type]['Dim1'].value_counts().head(10)
    print(f"\n{dim_type}:")
    print(dim_values)
    print("-" * 40)

print("\n\n")
print("=" * 60)
print("DIM2 TYPES AND THEIR VALUES")
print("=" * 60)

for dim_type in df['Dim2Type'].dropna().unique():
    dim_values = df[df['Dim2Type'] == dim_type]['Dim2'].value_counts().head(10)
    print(f"\n{dim_type}:")
    print(dim_values)
    print("-" * 40)
```

=====

DIM1 TYPES AND THEIR VALUES

=====

AMRGLASSCATEGORY:

Dim1

|                                   |    |
|-----------------------------------|----|
| AMRGLASSCATEGORY_AMRGLASS_UNKNOWN | 12 |
|-----------------------------------|----|

|                                            |    |
|--------------------------------------------|----|
| AMRGLASSCATEGORY_AMRGLASS_PROVIDEDTOLABS06 | 10 |
|--------------------------------------------|----|

|                                            |   |
|--------------------------------------------|---|
| AMRGLASSCATEGORY_AMRGLASS_PROVIDEDTOLABS07 | 2 |
|--------------------------------------------|---|

Name: count, dtype: int64

-----

ASSISTIVETECHSUBQUESTION:

Dim1

|                                               |    |
|-----------------------------------------------|----|
| ASSISTIVETECHSUBQUESTION_ASSISTIVETECH_Q07_02 | 70 |
|-----------------------------------------------|----|

|                                               |    |
|-----------------------------------------------|----|
| ASSISTIVETECHSUBQUESTION_ASSISTIVETECH_Q07_06 | 70 |
|-----------------------------------------------|----|

|                                               |    |
|-----------------------------------------------|----|
| ASSISTIVETECHSUBQUESTION_ASSISTIVETECH_Q07_04 | 70 |
|-----------------------------------------------|----|

|                                               |    |
|-----------------------------------------------|----|
| ASSISTIVETECHSUBQUESTION_ASSISTIVETECH_Q07_01 | 70 |
|-----------------------------------------------|----|

|                                               |    |
|-----------------------------------------------|----|
| ASSISTIVETECHSUBQUESTION_ASSISTIVETECH_Q07_05 | 70 |
|-----------------------------------------------|----|

|                                               |    |
|-----------------------------------------------|----|
| ASSISTIVETECHSUBQUESTION_ASSISTIVETECH_Q07_03 | 70 |
|-----------------------------------------------|----|

|                                                |    |
|------------------------------------------------|----|
| ASSISTIVETECHSUBQUESTION_ASSISTIVETECH_SUMMARY | 70 |
|------------------------------------------------|----|

Name: count, dtype: int64

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ASSISTIVETECHPRODUCT:

Dim1

|                                          |     |
|------------------------------------------|-----|
| ASSISTIVETECHPRODUCT_ASSISTIVETECH_TOTAL | 156 |
|------------------------------------------|-----|

|                                                  |     |
|--------------------------------------------------|-----|
| ASSISTIVETECHPRODUCT_ASSISTIVETECH_TOTAL_NOSPECS | 156 |
|--------------------------------------------------|-----|

Name: count, dtype: int64

-----

SEX:

Dim1

|          |        |
|----------|--------|
| SEX_FMLE | 907147 |
|----------|--------|

|         |        |
|---------|--------|
| SEX_MLE | 905115 |
|---------|--------|

|          |        |
|----------|--------|
| SEX_BTSX | 602371 |
|----------|--------|

Name: count, dtype: int64

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EMFEXPOSED:

Dim1

|                      |     |
|----------------------|-----|
| EMFEXPOSED_EMFWORKER | 117 |
|----------------------|-----|

|                      |     |
|----------------------|-----|
| EMFEXPOSED_EMFPUBLIC | 115 |
|----------------------|-----|

Name: count, dtype: int64

-----

#### WEALTHQUINTILE:

Dim1

|                     |      |
|---------------------|------|
| WEALTHQUINTILE_WQ3  | 4585 |
| WEALTHQUINTILE_WQ2  | 4585 |
| WEALTHQUINTILE_WQ1  | 4585 |
| WEALTHQUINTILE_WQ4  | 4585 |
| WEALTHQUINTILE_WQ5  | 4584 |
| WEALTHQUINTILE_TOTL | 2291 |

Name: count, dtype: int64

#### RESIDENCEAREATYPE:

Dim1

|                        |        |
|------------------------|--------|
| RESIDENCEAREATYPE_TOTL | 131667 |
| RESIDENCEAREATYPE_URB  | 127578 |
| RESIDENCEAREATYPE_RUR  | 126269 |

Name: count, dtype: int64

#### HOUSEHOLD\_COMP\_BY\_AGE:

Dim1

|                                                   |      |
|---------------------------------------------------|------|
| HOUSEHOLD_COMP_BY_AGE_HH_COMP_MULTIGEN            | 1184 |
| HOUSEHOLD_COMP_BY_AGE_HH_COMP_ADULTS_WITH_KIDS    | 1151 |
| HOUSEHOLD_COMP_BY_AGE_HH_COMP_ADULTS              | 1129 |
| HOUSEHOLD_COMP_BY_AGE_HH_COMP_ADULTS_WITH_OLDER   | 1124 |
| HOUSEHOLD_COMP_BY_AGE_HH_COMP_MULTIGEN_WITH_OLDER | 1105 |
| HOUSEHOLD_COMP_BY_AGE_HH_COMP_ONLY_OLDER_ADULTS   | 1062 |
| HOUSEHOLD_COMP_BY_AGE_HH_COMP_ONLY_BELOW_20_YEARS | 147  |

Name: count, dtype: int64

#### AGEGROUP:

Dim1

|                      |      |
|----------------------|------|
| AGEGROUP_YEARS20-49  | 3456 |
| AGEGROUP_YEARS15-19  | 3456 |
| AGEGROUP_YEARSALL    | 1668 |
| AGEGROUP_AGE60+      | 1451 |
| AGEGROUP_AGE160      | 1451 |
| AGEGROUP_MONTHS6-11  | 990  |
| AGEGROUP_MONTHS12-23 | 990  |
| AGEGROUP_MONTHS24-35 | 765  |
| AGEGROUP_MONTHS36-47 | 759  |
| AGEGROUP_MONTHS48-59 | 751  |

Name: count, dtype: int64

-----

GOEQUESTION:

Dim1

|                            |     |
|----------------------------|-----|
| GOEQUESTION_GOEQUESTION016 | 366 |
| GOEQUESTION_GOEQUESTION014 | 242 |
| GOEQUESTION_GOEQUESTION009 | 123 |
| GOEQUESTION_GOEQUESTION013 | 123 |
| GOEQUESTION_GOEQUESTION011 | 123 |
| GOEQUESTION_GOEQUESTION012 | 123 |
| GOEQUESTION_GOEQUESTION015 | 123 |

Name: count, dtype: int64

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TOBACCO\_INDICATOR:

Dim1

|                                       |      |
|---------------------------------------|------|
| TOBACCO_INDICATOR_PRICE_IN_CURRENCY   | 3705 |
| TOBACCO_INDICATOR_PRICE_IN_PPP        | 3705 |
| TOBACCO_INDICATOR_CURRENCY_LOCAL      | 3705 |
| TOBACCO_INDICATOR_PRICE_IN_USD        | 3705 |
| TOBACCO_INDICATOR_TAX_EXCISEPROPPRICE | 2925 |
| TOBACCO_INDICATOR_TAX_VALUEADDED      | 2925 |
| TOBACCO_INDICATOR_TAX_IMPORTDUTIES    | 2925 |
| TOBACCO_INDICATOR_TAX_TOTALTAX        | 2925 |
| TOBACCO_INDICATOR_TAX_SPECIFICEXCISE  | 2925 |
| TOBACCO_INDICATOR_TAX_ADVALEXCISE     | 2925 |

Name: count, dtype: int64

-----

TOBACCO\_NICOTINE\_PRODUCT:

Dim1

|                                                               |      |
|---------------------------------------------------------------|------|
| TOBACCO_NICOTINE_PRODUCT_CIGARETTE_MOSTSOLD                   | 6240 |
| TOBACCO_NICOTINE_PRODUCT_TOBACCO_OTHERSMOKED_MOSTSOLD         | 1170 |
| TOBACCO_NICOTINE_PRODUCT_TOBACCO_SMOKELESS_MOSTSOLD           | 975  |
| TOBACCO_NICOTINE_PRODUCT_CIGARETTE_PREMIUM                    | 780  |
| TOBACCO_NICOTINE_PRODUCT_ELILQUID_CLOSED_RECHARGABLE_CHEAPEST | 780  |
| TOBACCO_NICOTINE_PRODUCT_TOBACCO_HEATED_HTP_MOSTSOLD          | 780  |
| TOBACCO_NICOTINE_PRODUCT_CIGARETTE_CHEAPEST                   | 780  |
| TOBACCO_NICOTINE_PRODUCT_ELILQUID_OPEN_CHEAPEST               | 780  |
| TOBACCO_NICOTINE_PRODUCT_ELILQUID_CLOSED_DISPOSABLE_CHEAPEST  | 780  |

Name: count, dtype: int64

-----

PRICE:

Dim1

PRICE\_PRICE\_IN\_PPP 5  
Name: count, dtype: int64

-----  
DHSMICSGEOREGION:

Dim1

|                                |   |
|--------------------------------|---|
| DHSMICSGEOREGION_KHMDHS200014  | 9 |
| DHSMICSGEOREGION_PERDHS200413  | 9 |
| DHSMICSGEOREGION_RWADHS201002  | 9 |
| DHSMICSGEOREGION_GMBDHS201902  | 9 |
| DHSMICSGEOREGION_PERDHS200913  | 9 |
| DHSMICSGEOREGION_NAMDHS200004  | 9 |
| DHSMICSGEOREGION_IRQMICS201805 | 9 |
| DHSMICSGEOREGION_SENDHS201612  | 9 |
| DHSMICSGEOREGION_VUTMICS200705 | 9 |
| DHSMICSGEOREGION_CODDHS201304  | 9 |

Name: count, dtype: int64  
-----

WEALTHDECILE:

Dim1

|                             |      |
|-----------------------------|------|
| WEALTHDECILE_WEALTHDECILE09 | 3932 |
| WEALTHDECILE_WEALTHDECILE08 | 3932 |
| WEALTHDECILE_WEALTHDECILE03 | 3932 |
| WEALTHDECILE_WEALTHDECILE10 | 3932 |
| WEALTHDECILE_WEALTHDECILE06 | 3932 |
| WEALTHDECILE_WEALTHDECILE02 | 3932 |
| WEALTHDECILE_WEALTHDECILE01 | 3931 |
| WEALTHDECILE_WEALTHDECILE04 | 3931 |
| WEALTHDECILE_WEALTHDECILE07 | 3931 |
| WEALTHDECILE_WEALTHDECILE05 | 3930 |

Name: count, dtype: int64  
-----

EDUCATIONLEVEL:

Dim1

|                                  |      |
|----------------------------------|------|
| EDUCATIONLEVEL_SHLM              | 4552 |
| EDUCATIONLEVEL_PRLM              | 4498 |
| EDUCATIONLEVEL_NHLM              | 4439 |
| EDUCATIONLEVEL_TOTL              | 1271 |
| EDUCATIONLEVEL_EDUC_NONE_PRIMARY | 632  |
| EDUCATIONLEVEL_EDUC_SECONDARY    | 596  |
| EDUCATIONLEVEL_EDUC_HIGHER       | 507  |
| EDUCATIONLEVEL_MASTER            | 388  |
| EDUCATIONLEVEL_BACHELOR          | 388  |



EDUCATIONLEVEL\_DOCTORAL 388

Name: count, dtype: int64

SEVERITY:

Dim1

SEVERITY\_TOTAL 4040

SEVERITY\_MILD 4040

SEVERITY\_MODERATE 4040

SEVERITY\_SEVERE 4040

Name: count, dtype: int64

HOUSEHOLDWEALTH:

Dim1

HOUSEHOLDWEALTH\_TOTL 1271

HOUSEHOLDWEALTH\_WEALTH\_TOP40 638

HOUSEHOLDWEALTH\_WEALTH\_TOP60 638

HOUSEHOLDWEALTH\_WEALTH\_BOTTOM60 638

HOUSEHOLDWEALTH\_WEALTH\_BOTTOM80 638

HOUSEHOLDWEALTH\_WEALTH\_TOP80 638

HOUSEHOLDWEALTH\_WEALTH\_BOTTOM40 638

HOUSEHOLDWEALTH\_WEALTH\_MIDDLE60 419

Name: count, dtype: int64

WEALTHTERCILE:

Dim1

WEALTHTERCILE\_TOTL 1036

WEALTHTERCILE\_WEALTHTERCILE2 4

WEALTHTERCILE\_WEALTHTERCILE1 4

WEALTHTERCILE\_WEALTHTERCILE3 4

Name: count, dtype: int64

SUBSTANCETYPEDISORDER:

Dim1

SUBSTANCETYPEDISORDER\_ALCOHOL\_USE\_DISORDERS 512

SUBSTANCETYPEDISORDER\_DRUG\_USE\_DISORDERS 510

Name: count, dtype: int64

RSUDTREATMENTLOCATION:

Dim1

RSUDTREATMENTLOCATION\_IN\_THE\_CAPITAL\_CITY 1552

```
RSUDTREATMENTLOCATION_IN_OTHER_AREAS      1552
RSUDTREATMENTLOCATION_IN_OTHER_MAJOR_CITIES 1552
Name: count, dtype: int64
```

GROUP:

```
Dim1
GROUP_AL_ANON_ALATEEN      194
GROUP_COCAINE_ANONYMOUS    194
GROUP_FAMILY_SUPPORT_GROUPS 194
GROUP_NARCOTICS_ANONYMOUS  194
GROUP_ALCOHOLICS_ANONYMOUS 194
Name: count, dtype: int64
```

LEGISLATIONTYPE:

```
Dim1
LEGISLATIONTYPE_SA_ILLEGAL_PRODUCTION    45
LEGISLATIONTYPE_SA_ILLEGAL_SALES         10
Name: count, dtype: int64
```

ADVERTISINGTYPE:

```
Dim1
ADVERTISINGTYPE_SA_WINE_ADS      542
ADVERTISINGTYPE_SA_BEER_ADS      542
ADVERTISINGTYPE_SA_SPIRITS_ADS   542
Name: count, dtype: int64
```

BEVERAGETYPE:

```
Dim1
BEVERAGETYPE_ALCOHOLIC      1950
BEVERAGETYPE_NONALCOHOLIC   390
BEVERAGETYPE_SA_750_POPULAR_IMPORTED_SPIRITS 380
BEVERAGETYPE_SA_500_POPULAR_BEER             378
BEVERAGETYPE_SA_750_TABLE_WINE               378
BEVERAGETYPE_SA_750_POPULAR_LOCAL_SPIRITS    378
Name: count, dtype: int64
```

ALCOHOLTYPE:

```
Dim1
ALCOHOLTYPE_SA_BEER      31196
ALCOHOLTYPE_SA_SPIRITS   31191
```

|                              |       |
|------------------------------|-------|
| ALCOHOLTYPE_SA_WINE          | 31188 |
| ALCOHOLTYPE_SA_TOTAL         | 22754 |
| ALCOHOLTYPE_SA_OTHER_ALCOHOL | 21955 |

Name: count, dtype: int64

SCSUDSURVEYQUESTIONS:

Dim1

|                                                                        |     |
|------------------------------------------------------------------------|-----|
| SCSUDSURVEYQUESTIONS_SCSUD41_07_independentbody_licensing_health_profs | 145 |
| SCSUDSURVEYQUESTIONS_SCSUD41_05_contprofdev_community_hlthwrkr         | 145 |
| SCSUDSURVEYQUESTIONS_SCSUD41_06_attain_masters_or_equivalent_level     | 145 |
| SCSUDSURVEYQUESTIONS_SCSUD41_05_postgradtrainprogs_field_workers       | 145 |
| SCSUDSURVEYQUESTIONS_SCSUD41_05_postgradtrainprogs_nurses              | 145 |
| SCSUDSURVEYQUESTIONS_SCSUD41_05_contprofdev_nurses                     | 145 |
| SCSUDSURVEYQUESTIONS_SCSUD41_05_postgradtrainprogs_psychiatrists       | 145 |
| SCSUDSURVEYQUESTIONS_SCSUD41_05_postgradtrainprogs_addict_counsellors  | 145 |
| SCSUDSURVEYQUESTIONS_SCSUD41_05_contprofdev_narcologists               | 145 |
| SCSUDSURVEYQUESTIONS_SCSUD41_06_attain_short_cycle_tertiary_education  | 145 |

Name: count, dtype: int64

TB\_TREATMENTTYPE:

Dim1

|                                        |    |
|----------------------------------------|----|
| TB_TREATMENTTYPE_TB_TREATMENTTYPE_DRTB | 36 |
| TB_TREATMENTTYPE_TB_TREATMENTTYPE_DSTB | 36 |
| TB_TREATMENTTYPE_TB_TREATMENTTYPE_TOTL | 20 |

Name: count, dtype: int64

TB\_TREATHISTORY:

Dim1

|                                           |     |
|-------------------------------------------|-----|
| TB_TREATHISTORY_TB_TREATHISTORY_NEW       | 959 |
| TB_TREATHISTORY_TB_TREATHISTORY_RETREATED | 940 |

Name: count, dtype: int64

DRIVERTYPE:

Dim1

|                                  |     |
|----------------------------------|-----|
| DRIVERTYPE_RS-DRIVERTYPE-YOUNG   | 175 |
| DRIVERTYPE_RS-DRIVERTYPE-GENERAL | 175 |

Name: count, dtype: int64

RSUDREP:

Dim1

|                                       |     |
|---------------------------------------|-----|
| RSUDREP_RSUDREP_NGO                   | 776 |
| RSUDREP_RSUDREP_TOBACCOINDUSTRY       | 776 |
| RSUDREP_RSUDREP_ALCOHOLUSEDISORDERFAM | 776 |
| RSUDREP_RSUDREP_DRUGUSER              | 776 |
| RSUDREP_RSUDREP_DRUGUSEDISORDER       | 776 |
| RSUDREP_RSUDREP_ALCOHOLINDUSTRY       | 776 |
| RSUDREP_RSUDREP_PHARMAINDUSTRY        | 776 |
| RSUDREP_RSUDREP_DRUGUSEDISORDERFAM    | 776 |
| RSUDREP_RSUDREP_YOUTH                 | 776 |
| RSUDREP_RSUDREP_ALCOHOLUSEDISORDER    | 776 |

Name: count, dtype: int64

#### RSUDTREATMENT:

Dim1

|                                         |     |
|-----------------------------------------|-----|
| RSUDTREATMENT_RSUDTREATMENT_ALTERNATIVE | 388 |
| RSUDTREATMENT_RSUDTREATMENT_ADDITION    | 388 |

Name: count, dtype: int64

#### GOVERNMENTBENEFIT:

Dim1

|                                        |     |
|----------------------------------------|-----|
| GOVERNMENTBENEFIT_NON_MONETARY_SUPPORT | 194 |
| GOVERNMENTBENEFIT_MONETARY_SUPPORT     | 194 |
| GOVERNMENTBENEFIT_DISABILITY_PENSION   | 55  |
| GOVERNMENTBENEFIT_SUBSIDY_FOOD_HOUSING | 55  |

Name: count, dtype: int64

#### OPENACCESSSERVICE:

Dim1

|                                                    |     |
|----------------------------------------------------|-----|
| OPENACCESSSERVICE_WEB_BASED_INTERVENTIONS          | 194 |
| OPENACCESSSERVICE_MOBILE_PHONE_BASED_INTERVENTIONS | 194 |
| OPENACCESSSERVICE_TELEPHONE_HELP_LINES             | 194 |

Name: count, dtype: int64

#### RSUDHWF:

Dim1

|                                                       |     |
|-------------------------------------------------------|-----|
| RSUDHWF_ADDICTION_COUNSELLORS                         | 388 |
| RSUDHWF_DOCTORS_NOT_SPECIALIZED_IN_PSYCH_OR_ADDICTMED | 388 |
| RSUDHWF_PSYCHIATRIC_NURSES                            | 388 |
| RSUDHWF_SOCIAL_WORKERS                                | 388 |
| RSUDHWF_OUTREACH_FIELD_WORKERS                        | 388 |
| RSUDHWF_PSYCHIATRISTS                                 | 388 |

|                                         |     |
|-----------------------------------------|-----|
| RSUDHWF_COMMUNITY_HEALTH_WORKERS        | 388 |
| RSUDHWF_NURSES_NOT_SPECIALIZED_IN_PSYCH | 388 |
| RSUDHWF_NARCOLOGISTS                    | 388 |
| RSUDHWF_PSYCHOLOGISTS                   | 388 |

Name: count, dtype: int64

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PROGRAMME:

|                                      |     |
|--------------------------------------|-----|
| Dim1                                 |     |
| PROGRAMME_IN_PATIENT_TREATMENT       | 388 |
| PROGRAMME_OUT_PATIENT_TREATMENT      | 388 |
| PROGRAMME_OUT_PATIENT_DETOXIFICATION | 388 |
| PROGRAMME_REHABILITATION             | 388 |
| PROGRAMME_IN_PATIENT_DETOXIFICATION  | 388 |

Name: count, dtype: int64

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NGO:

|                    |     |
|--------------------|-----|
| Dim1               |     |
| NGO_PREVENTION     | 137 |
| NGO_REHABILITATION | 136 |
| NGO_TREATMENT      | 136 |
| NGO_ADVOCACY       | 135 |

Name: count, dtype: int64

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RSUDMONITORING:

|                                 |     |
|---------------------------------|-----|
| Dim1                            |     |
| RSUDMONITORING_FORENSIC_EXAMS   | 194 |
| RSUDMONITORING_TOXICOLOGY_UNITS | 194 |

Name: count, dtype: int64

-----

DRUG:

|                         |     |
|-------------------------|-----|
| Dim1                    |     |
| DRUG_ACAMPROSATE        | 318 |
| DRUG_DISULFIRAM         | 194 |
| DRUG_NALTREXONE         | 194 |
| DRUG_DIAZEPAM           | 194 |
| DRUG_BENZODIAZEPINES    | 131 |
| DRUG_CHLORPROMAZINE     | 126 |
| DRUG_NEW_ANTIPSYCHOTICS | 125 |
| DRUG_ALCOHOL_INFUSION   | 121 |
| DRUG_TIAGABINE          | 120 |
| DRUG_NITROUS_OXIDE      | 120 |

Name: count, dtype: int64

-----

SPECIALPOPULATION:

Dim1

|                                               |     |
|-----------------------------------------------|-----|
| SPECIALPOPULATION_CHILDREN_AND_FAMILIES_RISK  | 145 |
| SPECIALPOPULATION_CHILDREN_AND_ADOLESCENTS    | 145 |
| SPECIALPOPULATION_COMMERCIAL_SEX_WORKERS      | 145 |
| SPECIALPOPULATION_REFUGEES                    | 145 |
| SPECIALPOPULATION_PEOPLE_LIVING_WITH_HIV_AIDS | 145 |
| SPECIALPOPULATION_MINORITY_GROUPS             | 145 |
| SPECIALPOPULATION_PREGNANT_WOMEN              | 145 |
| SPECIALPOPULATION_PRISONERS                   | 145 |
| SPECIALPOPULATION_INDIGENOUS_POPULATIONS      | 145 |

Name: count, dtype: int64

-----

RSUDSPECIFICPOPULATION:

Dim1

|                                                             |     |
|-------------------------------------------------------------|-----|
| RSUDSPECIFICPOPULATION_CHILDREN_AND_FAMILIES_RISK           | 388 |
| RSUDSPECIFICPOPULATION_REFUGEES                             | 388 |
| RSUDSPECIFICPOPULATION_COMMERCIAL_SEX_WORKERS               | 388 |
| RSUDSPECIFICPOPULATION_CHILDREN_AND_ADOLESCENTS             | 388 |
| RSUDSPECIFICPOPULATION_YOUNG_ADULTS                         | 388 |
| RSUDSPECIFICPOPULATION_PARENTS                              | 388 |
| RSUDSPECIFICPOPULATION_INDIGENOUS_POPULATIONS               | 388 |
| RSUDSPECIFICPOPULATION_PERSONS_WITH_MENTAL_HEALTH_DISORDERS | 388 |
| RSUDSPECIFICPOPULATION_PREGNANT_WOMEN                       | 388 |
| RSUDSPECIFICPOPULATION_PEOPLE_WITH_TB                       | 388 |

Name: count, dtype: int64

-----

RSUDPREVENTIONPROGRAMME:

Dim1

|                                                     |     |
|-----------------------------------------------------|-----|
| RSUDPREVENTIONPROGRAMME_MASS_MEDIA_PRINT            | 194 |
| RSUDPREVENTIONPROGRAMME_SCHOOL_MANUAL_BASED         | 194 |
| RSUDPREVENTIONPROGRAMME_ADS_IN_PUBLIC               | 194 |
| RSUDPREVENTIONPROGRAMME_MASS_MEDIA_AUDIOVISUAL      | 194 |
| RSUDPREVENTIONPROGRAMME_WORK_PLACE_PROG             | 194 |
| RSUDPREVENTIONPROGRAMME_COMMUNITY_BASED_PROG        | 194 |
| RSUDPREVENTIONPROGRAMME_PARENT_PROG                 | 194 |
| RSUDPREVENTIONPROGRAMME_SERVING_STAFF_TRAINING_PROG | 194 |

Name: count, dtype: int64

-----

#### CONSUMPTIONTYPE:

Dim1

|                                        |    |
|----------------------------------------|----|
| CONSUMPTIONTYPE_CONSUMPTION_UNRECORDED | 42 |
| CONSUMPTIONTYPE_CONSUMPTION_RECORDED   | 42 |
| CONSUMPTIONTYPE_CONSUMPTION_TOTAL      | 42 |

Name: count, dtype: int64

#### ARCHIVE:

Dim1

ARCHIVE\_DEC2019 1153

Name: count, dtype: int64

#### SOCIALCOSTTYPE:

Dim1

|                                       |    |
|---------------------------------------|----|
| SOCIALCOSTTYPE_SA_DIRECT_OTHER_COSTS  | 22 |
| SOCIALCOSTTYPE_SA_INDIRECT_COSTS      | 22 |
| SOCIALCOSTTYPE_SA_DIRECT_HEALTH_COSTS | 22 |
| SOCIALCOSTTYPE_SA_TOTAL_COSTS         | 22 |
| SOCIALCOSTTYPE_SA_DIRECT_LAW_COSTS    | 20 |

Name: count, dtype: int64

#### HARMANDCONSEQUENCE:

Dim1

|                                                  |    |
|--------------------------------------------------|----|
| HARMANDCONSEQUENCE_SA_ALCOHOL_RELATD_SOCL_CNSQNC | 92 |
| HARMANDCONSEQUENCE_SA_ALCOHOL_RELATD_HLTH_CNSQNC | 92 |
| HARMANDCONSEQUENCE_SA_ALCOHOL_RELATED_HARM       | 91 |

Name: count, dtype: int64

#### BACGROUP:

Dim1

|                                      |     |
|--------------------------------------|-----|
| BACGROUP_SA_GENERAL_POPULATION       | 348 |
| BACGROUP_SA_YOUNG_NOVICE_DRIVERS     | 348 |
| BACGROUP_SA_PROFESS_COMMERCE_DRIVERS | 348 |

Name: count, dtype: int64

#### NATIONALSYSTEMTYPE:

Dim1

|                                                  |     |
|--------------------------------------------------|-----|
| NATIONALSYSTEMTYPE_SA_NST_SOCIALCONSEQUENCES     | 257 |
| NATIONALSYSTEMTYPE_SA_NST_ALCOHOLPOLICYRESPONSES | 218 |
| NATIONALSYSTEMTYPE_SA_NST_HEALTHCONSEQUENCES     | 208 |

Name: count, dtype: int64

-----

POLICYFRAMEWORK:

Dim1

|                                            |    |
|--------------------------------------------|----|
| POLICYFRAMEWORK_INTEGRATED_SUBSTANCE_ABUSE | 85 |
| POLICYFRAMEWORK_INTEGRATED_GEN_PUB_HEALTH  | 85 |
| POLICYFRAMEWORK_INTEGRATED_MENTAL_HEALTH   | 84 |
| POLICYFRAMEWORK_INTEGRATED_OTHER           | 82 |
| POLICYFRAMEWORK_INTEGRATED_NONCOM_DISEASES | 81 |
| POLICYFRAMEWORK_SPECIFIC_ALCOHOL_POLICY    | 81 |

Name: count, dtype: int64

-----

PUBLICPLACE:

Dim1

|                                           |     |
|-------------------------------------------|-----|
| PUBLICPLACE_SA_LEISURE_EVENTS             | 260 |
| PUBLICPLACE_SA_HEALTH_CARE_ESTABLISHMENTS | 260 |
| PUBLICPLACE_SA_GOVERNMENT_OFFICES         | 260 |
| PUBLICPLACE_SA_PARKS_AND_STREETS          | 260 |
| PUBLICPLACE_SA_EDUCATIONAL_BUILDINGS      | 260 |
| PUBLICPLACE_SA_SPORTING_EVENTS            | 258 |
| PUBLICPLACE_SA_PUBLIC_TRANSPORT           | 256 |
| PUBLICPLACE_SA_RELIGIOUS_PLACES           | 249 |
| PUBLICPLACE_SA_WORKPLACES                 | 241 |

Name: count, dtype: int64

-----

COORDINATINGENTITIES:

Dim1

|                                                |    |
|------------------------------------------------|----|
| COORDINATINGENTITIES_OTHER_COORDINATING_ENTITY | 86 |
| COORDINATINGENTITIES_AGRICULTURE               | 85 |
| COORDINATINGENTITIES_FINANCE_TAXATION          | 85 |
| COORDINATINGENTITIES_HEALTH                    | 84 |
| COORDINATINGENTITIES_EDUCATION                 | 83 |
| COORDINATINGENTITIES_ROAD_SAFETY               | 83 |
| COORDINATINGENTITIES_CRIMINAL_JUSTICE          | 83 |
| COORDINATINGENTITIES_TRADE_INDUSTRY            | 82 |
| COORDINATINGENTITIES_SOCIAL_AFFAIRS            | 82 |
| COORDINATINGENTITIES_EMPLOYMENT                | 82 |

Name: count, dtype: int64

-----

POLICYSECTOR:



Dim1

|                               |     |
|-------------------------------|-----|
| POLICYSECTOR_TRADE_INDUSTRY   | 121 |
| POLICYSECTOR_EDUCATION        | 115 |
| POLICYSECTOR_HEALTH_SECTOR    | 115 |
| POLICYSECTOR_FINANCE_TAXATION | 115 |
| POLICYSECTOR_EMPLOYMENT       | 115 |
| POLICYSECTOR_LAW_ENFORCEMENT  | 114 |
| POLICYSECTOR_AGRICULTURE      | 111 |
| POLICYSECTOR_JUSTICE          | 110 |
| POLICYSECTOR_SOCIAL           | 109 |
| POLICYSECTOR_ROAD_SAFETY      | 109 |

Name: count, dtype: int64

-----

#### ENVCAUSE:

Dim1

|                      |     |
|----------------------|-----|
| ENVCAUSE_ENVCAUSE001 | 388 |
| ENVCAUSE_ENVCAUSE000 | 388 |
| ENVCAUSE_ENVCAUSE151 | 388 |
| ENVCAUSE_ENVCAUSE060 | 388 |

Name: count, dtype: int64

-----

#### POLICYADOPTIONLEVEL:

Dim1

|                                          |    |
|------------------------------------------|----|
| POLICYADOPTIONLEVEL_OTHER_ADOPTION_LEVEL | 16 |
| POLICYADOPTIONLEVEL_SPECIFIC_MINISTRY    | 11 |
| POLICYADOPTIONLEVEL_NATIONAL_GOVERNMENT  | 10 |
| POLICYADOPTIONLEVEL_NATIONAL_PARLIAMENT  | 10 |

Name: count, dtype: int64

-----

=====

DIM2 TYPES AND THEIR VALUES

=====

#### ASSISTIVETECHSOURCE:

Dim2

|                                             |    |
|---------------------------------------------|----|
| ASSISTIVETECHSOURCE_ASSISTIVETECH_Private   | 52 |
| ASSISTIVETECHSOURCE_ASSISTIVETECH_NGO       | 52 |
| ASSISTIVETECHSOURCE_ASSISTIVETECH_Self-made | 52 |
| ASSISTIVETECHSOURCE_ASSISTIVETECH_Other     | 52 |
| ASSISTIVETECHSOURCE_ASSISTIVETECH_Public    | 52 |

Name: count, dtype: int64

-----

#### AGEGROUP:

Dim2

|                       |        |
|-----------------------|--------|
| AGEGROUP_YEARS18-PLUS | 305859 |
| AGEGROUP_AGEAll       | 136244 |
| AGEGROUP_YEARS10-19   | 113724 |
| AGEGROUP_YEARS30-49   | 89661  |
| AGEGROUP_YEARS15-29   | 89661  |
| AGEGROUP_YEARS50-69   | 89112  |
| AGEGROUP_YEARS70+     | 89112  |
| AGEGROUP_YEARS05-14   | 89112  |
| AGEGROUP_YEARS05-09   | 88326  |
| AGEGROUP_YEARS05-19   | 88326  |

Name: count, dtype: int64

-----

#### EMFFREQUENCY:

Dim2

|                        |     |
|------------------------|-----|
| EMFFREQUENCY_EMFLOW    | 183 |
| EMFFREQUENCY_EMFSTATIC | 49  |

Name: count, dtype: int64

-----

#### POP\_TYPE:

Dim2

|                                  |      |
|----------------------------------|------|
| POP_TYPE_POP_TYPE_HOUSEHOLD_HEAD | 5864 |
| POP_TYPE_CHILDREN                | 397  |
| POP_TYPE_GENERALPOPULATION       | 397  |

Name: count, dtype: int64

-----

#### PRICE:

Dim2

|                         |      |
|-------------------------|------|
| PRICE_PRICE_IN_USD      | 3120 |
| PRICE_CURRENCY          | 3120 |
| PRICE_PRICE_IN_PPP      | 3120 |
| PRICE_PRICE_IN_CURRENCY | 3120 |
| PRICE_QUANTITY          | 195  |

Name: count, dtype: int64

-----

#### HOUSEHOLDCOOKINGFUEL:

Dim2

|                                       |       |
|---------------------------------------|-------|
| HOUSEHOLDCOOKINGFUEL_FUEL_KEROSENE    | 34068 |
| HOUSEHOLDCOOKINGFUEL_FUEL_GAS         | 34068 |
| HOUSEHOLDCOOKINGFUEL_FUEL_COAL        | 34068 |
| HOUSEHOLDCOOKINGFUEL_FUEL_ELECTRICITY | 34068 |
| HOUSEHOLDCOOKINGFUEL_FUEL_CHARCOAL    | 34068 |
| HOUSEHOLDCOOKINGFUEL_FUEL_BIOMASS     | 34068 |

Name: count, dtype: int64

-----

BEVERAGE:

Dim2

|                               |     |
|-------------------------------|-----|
| BEVERAGE_BEV_BEER             | 975 |
| BEVERAGE_BEV_SPIRITS          | 975 |
| BEVERAGE_BEV_SUGAR_CARBONATED | 390 |

Name: count, dtype: int64

-----

TOBACCO\_NICOTINE\_PRODUCT:

Dim2

|                                                              |       |
|--------------------------------------------------------------|-------|
| TOBACCO_NICOTINE_PRODUCT_CIGARETTE_MOSTSOLD                  | 19695 |
| TOBACCO_NICOTINE_PRODUCT_TOBACCO_OTHERSMOKED_MOSTSOLD        | 2535  |
| TOBACCO_NICOTINE_PRODUCT_ELIQUID_CLOSED_DISPOSABLE_CHEAPEST  | 2340  |
| TOBACCO_NICOTINE_PRODUCT_ELIQUID_OPEN_CHEAPEST               | 2340  |
| TOBACCO_NICOTINE_PRODUCT_TOBACCO_SMOKELESS_MOSTSOLD          | 2340  |
| TOBACCO_NICOTINE_PRODUCT_ELIQUID_CLOSED_RECHARGABLE_CHEAPEST | 2340  |
| TOBACCO_NICOTINE_PRODUCT_TOBACCO_HEATED_HTP_MOSTSOLD         | 2145  |
| TOBACCO_NICOTINE_PRODUCT_CIGARETTE_PREMIUM                   | 1560  |
| TOBACCO_NICOTINE_PRODUCT_CIGARETTE_CHEAPEST                  | 1560  |

Name: count, dtype: int64

-----

ENVCAUSE:

Dim2

|                      |      |
|----------------------|------|
| ENVCAUSE_ENVCAUSE039 | 2727 |
| ENVCAUSE_ENVCAUSE000 | 2727 |
| ENVCAUSE_ENVCAUSE113 | 2727 |
| ENVCAUSE_ENVCAUSE114 | 2727 |
| ENVCAUSE_ENVCAUSE118 | 2727 |
| ENVCAUSE_ENVCAUSE068 | 2727 |
| ENVCAUSE_ENVCAUSE104 | 360  |

Name: count, dtype: int64

-----

GHECAUSE:

```

Dim2
GHECAUSE_GHE001180    22620
GHECAUSE_GHE000390    22620
GHECAUSE_GHE001130    22620
GHECAUSE_GHE000680    22620
GHECAUSE_GHE000000    22620
GHECAUSE_GHE001140    22620
Name: count, dtype: int64
-----

```

```

ARCHIVE:
Dim2
ARCHIVE_DEC2019    16652
Name: count, dtype: int64
-----

```

```

TB_CASETYPE:
Dim2
TB_CASETYPE_TB_CASETYPE_BC    207
TB_CASETYPE_TB_CASETYPE_SP    190
Name: count, dtype: int64
-----

```

```

In [57]: # Define aggregate/total values for each dimension type
aggregate_filters = {
    # Dim1 aggregates
    'SEX': ['SEX_BTSX'], # Both sexes
    'RESIDENCEAREATYPE': ['RESIDENCEAREATYPE_TOTL'], # Total (urban + rural)
    'AGEGROUP': ['AGEGROUP_YEARSALL'], # All ages
    'WEALTHQUINTILE': ['WEALTHQUINTILE_TOTL'], # Total wealth
    'EDUCATIONLEVEL': ['EDUCATIONLEVEL_TOTL'], # Total education
    'SEVERITY': ['SEVERITY_TOTAL'], # Total severity
    'HOUSEHOLDWEALTH': ['HOUSEHOLDWEALTH_TOTL'], # Total wealth
    'WEALTHTERCILE': ['WEALTHTERCILE_TOTL'], # Total wealth
    'ALCOHOLTYPE': ['ALCOHOLTYPE_SA_TOTAL'], # Total alcohol
    'TB_TREATMENTTYPE': ['TB_TREATMENTTYPE_TB_TREATMENTTYPE_TOTL'], # Total TB treatment
    'CONSUMPTIONTYPE': ['CONSUMPTIONTYPE_CONSUMPTION_TOTAL'], # Total consumption
    'ASSISTIVETECHPRODUCT': ['ASSISTIVETECHPRODUCT_ASSISTIVETECH_TOTAL'], # Total assistive tech

    # Dim2 aggregates
    'GHECAUSE': ['GHECAUSE_GHE000000'], # All causes
    'ENVCAUSE': ['ENVCAUSE_ENVCAUSE000'], # All environmental causes (likely)
}

# Also handle AGEGROUP in Dim2

```

```

dim2_agegroup_aggregates = ['AGEGROUP_AGEAll', 'AGEGROUP_YEARS18-PLUS']

df_filtered = df.copy()

# Filter Dim1 to aggregates
for dim_type, allowed_values in aggregate_filters.items():
    df_filtered = df_filtered[
        (df_filtered['Dim1Type'] != dim_type) | # Keep if not this dimension type
        (df_filtered['Dim1'].isin(allowed_values)) # Or keep if it's an aggregate value
    ]

# Filter Dim2 to aggregates
for dim_type, allowed_values in aggregate_filters.items():
    if dim_type == 'AGEGROUP':
        # Special handling for AGEGROUP in Dim2
        df_filtered = df_filtered[
            (df_filtered['Dim2Type'] != dim_type) |
            (df_filtered['Dim2'].isin(dim2_agegroup_aggregates))
        ]
    else:
        df_filtered = df_filtered[
            (df_filtered['Dim2Type'] != dim_type) |
            (df_filtered['Dim2'].isin(allowed_values))
        ]

# Count countries per indicator
indicator_coverage = df_filtered.groupby(['IndicatorCode',
    'IndicatorName'])['SpatialDim'].nunique().reset_index(name='Country_Count')
indicator_coverage = indicator_coverage.sort_values('Country_Count', ascending=False)

print(f"Total records before filtering: {len(df):,}")
print(f"Total records after filtering: {len(df_filtered):,}")
print(f"Reduction: {((len(df) - len(df_filtered)) / len(df) * 100):.1f}%")
print(f"\nNumber of unique indicators: {indicator_coverage.shape[0]}")
print(f"\nTop 200 indicators by country coverage:")

# Remove BMI indicators and add only NCD_BMI_30A indicator
top200Indicators = indicator_coverage[~indicator_coverage['IndicatorCode'].str.lower().str.contains('bmi')].head(200)
top200Indicators = pd.concat([indicator_coverage[indicator_coverage['IndicatorCode'] == 'NCD_BMI_30A'], top200Indicators]).reset_index

display(top200Indicators)

top200Indicators[['IndicatorCode', 'IndicatorName']].to_csv('../data/metadata/who_indicators_obesity_core.csv', index=False)

```

Total records before filtering: 3,961,866

Total records after filtering: 1,470,590

Reduction: 62.9%

Number of unique indicators: 924

Top 200 indicators by country coverage:

|     | IndicatorCode                          | IndicatorName                                     | Country_Count |
|-----|----------------------------------------|---------------------------------------------------|---------------|
| 0   | NCD_BMI_30A                            | Prevalence of obesity among adults, BMI &Great... | 210           |
| 1   | NUTRITION_ANAEMIA_CHILDREN_PREV        | Prevalence of anaemia in children aged 6-59 mo... | 242           |
| 2   | SDGIHR                                 | Average of 13 International Health Regulations... | 241           |
| 3   | SA_0000001688                          | Alcohol, total per capita (15+) consumption (i... | 235           |
| 4   | WHS3_46                                | Total tetanus - number of reported cases          | 224           |
| ... | ...                                    | ...                                               | ...           |
| 196 | R_admin_duty_free_banned               | Tobacco tax administration: Sales of duty (or ... | 195           |
| 197 | R_admin_identifier_track_and_trace     | Tobacco tax administration: Marking/unique ide... | 195           |
| 198 | R_admin_overt_marking                  | Tobacco tax administration: Marking is overt      | 195           |
| 199 | TOBACCO_MPOWER_M1_TOBACCOUSESMOKING    | Tobacco Monitor: M1 population-based national ... | 195           |
| 200 | TOBACCO_MPOWER_M2_SMOKELESSECIGARETTES | Tobacco Monitor: M2 population-based national ... | 195           |

201 rows × 3 columns

```
In [64]: cols = ['IndicatorCode', 'IndicatorName',  
                'SpatialDim', # Country code  
                'SpatialDimType',  
                'TimeDim', # Year  
                'Dim1Type', 'Dim1', # (e.g., SEX)  
                'Dim2Type', 'Dim2', # (e.g., AGEGROUP)  
                'NumericValue', 'Value', # The actual data  
                'Low', 'High' # Confidence intervals  
                ]  
  
df_tidy = df.copy()[cols]  
  
#filter columns
```

```

df_tidy = df[[
    'IndicatorCode', 'IndicatorName',
    'SpatialDim', 'TimeDim',
    'Dim1Type', 'Dim1',
    'Dim2Type', 'Dim2',
    'NumericValue', 'Value',
    'Low', 'High'
]].copy()

# Rename for clarity
df_tidy = df_tidy.rename(columns={
    'SpatialDim': 'Country',
    'TimeDim': 'Year',
    'NumericValue': 'Numeric_Value',
    'Value': 'Text_Value'
})

# Create a unified value column (choose numeric when available)
df_tidy['Measurement'] = df_tidy['Numeric_Value'].fillna(df_tidy['Text_Value'])

# Separate by Sex if Dim1Type is SEX
df_tidy['Sex'] = df_tidy.apply(
    lambda x: x['Dim1'] if x['Dim1Type'] == 'SEX' else 'Both',
    axis=1
)

# Separate by Age if Dim2Type is AGEGROUP
df_tidy['AgeGroup'] = df_tidy.apply(
    lambda x: x['Dim2'] if x['Dim2Type'] == 'AGEGROUP' else 'All',
    axis=1
)

# Drop the generic Dim columns
df_tidy = df_tidy.drop(columns=['Dim1Type', 'Dim1', 'Dim2Type', 'Dim2'])

# Final tidy structure for 2012
df_tidy = df_tidy[df_tidy['Year'] == 2012][[
    'IndicatorCode', 'IndicatorName',
    'Country', 'Year',
    'Sex', 'AgeGroup',
    'Measurement', 'Low', 'High'
]]

```

```

# Find indicators with strong correlation to NCD_BMI_30A

# Filter to numeric measurements only
df_analysis = df_tidy[pd.to_numeric(df_tidy['Measurement'],
errors='coerce').notna()].copy()
df_analysis['Measurement'] = pd.to_numeric(df_analysis['Measurement'])

# Create unique key for matching (Country + Year) - our focus is measurement per country and year
df_analysis['key'] = df_analysis['Country'] + '_' + df_analysis['Year'].astype(str)

# Pivot: Make indicators as columns, keep country-year as rows (Sort by highest measurement before pivoting)
df_analysis = df_analysis.sort_values('Measurement', ascending=False)
df_pivot = df_analysis.pivot_table(
    index='key',
    columns='IndicatorCode',
    values='Measurement',
    aggfunc='first' # Use first value if duplicates
)

print(f"Pivot shape: {df_pivot.shape}")
print(f"Number of country-year observations: {df_pivot.shape[0]}")
print(f"Number of indicators: {df_pivot.shape[1]}")

# Only calculate for indicators with sufficient data
min_observations = 180

correlations = []
bmi_data = df_pivot['NCD_BMI_30A']

for col in df_pivot.columns:
    if ('BMI' in col.upper()) : #skip BMI indicators
        continue

    # Count valid pairs
    valid_pairs = bmi_data.notna() & df_pivot[col].notna()
    n_pairs = valid_pairs.sum()

    if n_pairs >= min_observations:
        corr = bmi_data.corr(df_pivot[col])
        correlations.append({
            'IndicatorCode': col,
            'Correlation': corr,
            'Correlation-abs': abs(corr),

```



```

        'N_observations': n_pairs
    })

# Sort by correlation
corr_df = pd.DataFrame(correlations).sort_values('Correlation-abs', ascending=False)

# Filter for strong correlations (|correlation| > 0.4)
strong_corr = corr_df[corr_df['Correlation-abs'] > 0.4].copy()

print(f"\n✅ Found {len(strong_corr)} indicators with |correlation| > 0.4")
print(f"    (from {len(corr_df)} total indicators with sufficient data)")

# 8. Add indicator names
indicator_names = df_tidy[['IndicatorCode', 'IndicatorName']].drop_duplicates()
strong_corr = strong_corr.merge(indicator_names, on='IndicatorCode', how='left')

# Reorder columns
strong_corr = strong_corr[['IndicatorCode', 'IndicatorName', 'Correlation',
'N_observations']]

print("\n📊 Strongly correlated indicators (sorted by correlation):")
display(strong_corr)

strong_corr.to_csv('../data/metadata/indicators_correlated_with_obesity.csv', index=False)
print(f"\n💾 Saved to: data/metadata/indicators_correlated_with_obesity.csv")

```

Pivot shape: (345, 249)

Number of country-year observations: 345

Number of indicators: 249

✅ Found 14 indicators with |correlation| > 0.4  
 (from 97 total indicators with sufficient data)

📊 Strongly correlated indicators (sorted by correlation):

|    | IndicatorCode                  | IndicatorName                                           | Correlation | N_observations |
|----|--------------------------------|---------------------------------------------------------|-------------|----------------|
| 0  | NCD_GLUC_03                    | Raised fasting blood glucose ( $\geq 7.0$ mmol/L) (...) | 0.842245    | 201            |
| 1  | NCD_GLUC_04                    | Raised fasting blood glucose ( $\geq 7.0$ mmol/L) (a... | 0.830862    | 201            |
| 2  | NCD_DIABETES_PREVALENCE_CRUDE  | Prevalence of diabetes, crude                           | 0.760636    | 210            |
| 3  | NCD_DIABETES_PREVALENCE_AGESTD | Prevalence of diabetes, age-standardized                | 0.732298    | 210            |
| 4  | NCD_GLUC_01                    | Mean fasting blood glucose (mmol/l) (age-stand...       | 0.667667    | 192            |
| 5  | NCD_PAA                        | Prevalence of insufficient physical activity a...       | 0.491062    | 205            |
| 6  | NCD_PAC                        | Prevalence of insufficient physical activity a...       | 0.487815    | 205            |
| 7  | TOTENV_2                       | Age-standardized deaths attributable to the en...       | -0.477589   | 192            |
| 8  | PHE_HHAIR_PROP_POP_CLEAN_FUELS | Proportion of population with primary reliance...       | 0.471137    | 196            |
| 9  | TOTENV_4                       | Age-standardized DALYs attributable to the env...       | -0.458480   | 192            |
| 10 | WSH_SANITATION_BASIC           | Population using at least basic sanitation ser...       | 0.443851    | 200            |
| 11 | MCV2                           | Measles-containing-vaccine second-dose (MCV2) ...       | 0.417403    | 203            |
| 12 | AIR_52                         | Household air pollution attributable death rat...       | -0.412561   | 187            |
| 13 | WSH_WATER_BASIC                | Population using at least basic drinking-water...       | 0.411288    | 200            |

 Saved to: data/metadata/indicators\_correlated\_with\_obesity.csv